Software Engineering Lab - Lab 6

Code and Branch Coverage Analysis

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Assigned program - Income Tax Calculator

1. Objective

Measure statement and branch coverage of the assigned income tax module using pytest and Coverage.py. The application source is unchanged; both test suites exceed the required 90% coverage.

2. Statement Coverage

CodeCoverageTest_income_tax.py exercises all three functions: taxable_income(), tax_due() and refund_status(). It checks deductions, exemptions, invalid inputs and each refund status.

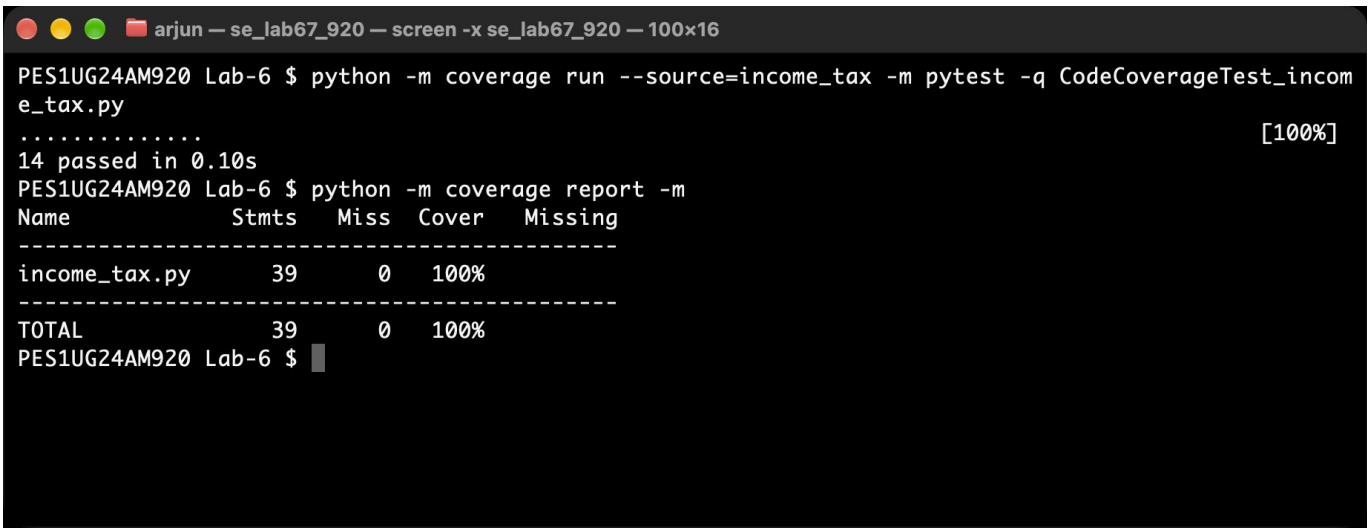


Figure 1. Statement coverage: 14 tests passed; all 39 statements executed.

3. Branch Coverage

BranchCoverageTest_income_tax.py adds decision outcomes and boundary cases for investment limits, age groups, exemptions, tax slabs and filing status.

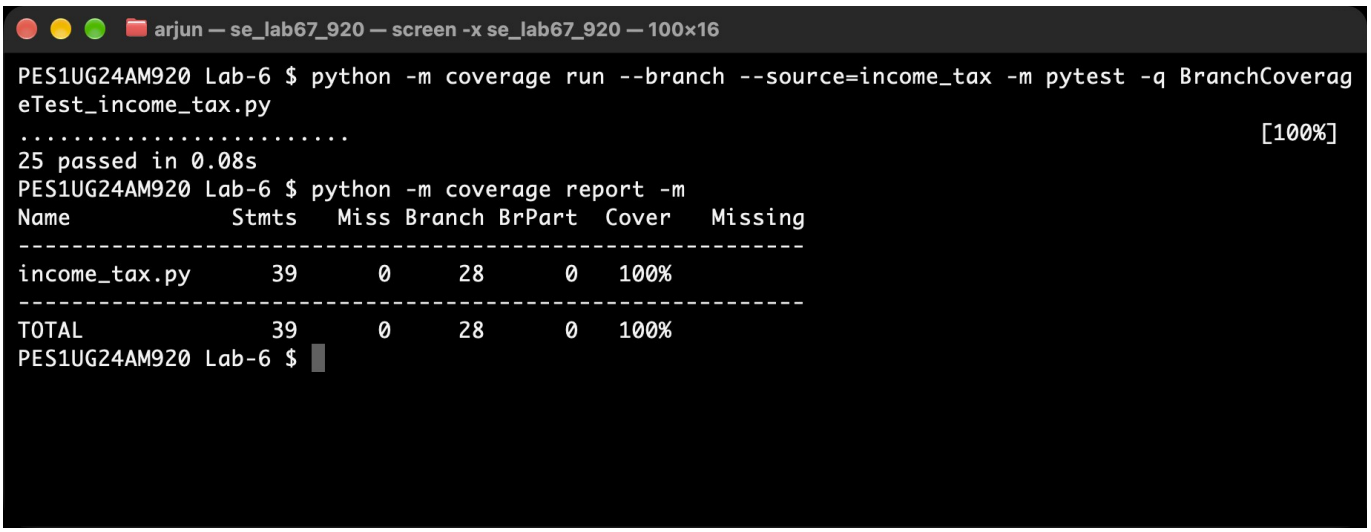


Figure 2. Branch coverage: 25 tests passed; all 39 statements and 28 branches executed.

4. Result

Both suites achieved 100% coverage with no missing statements or partial branches. The statement and branch test suites are recorded in separate commits.